YOLO Loss Functions

(for Different Versions)

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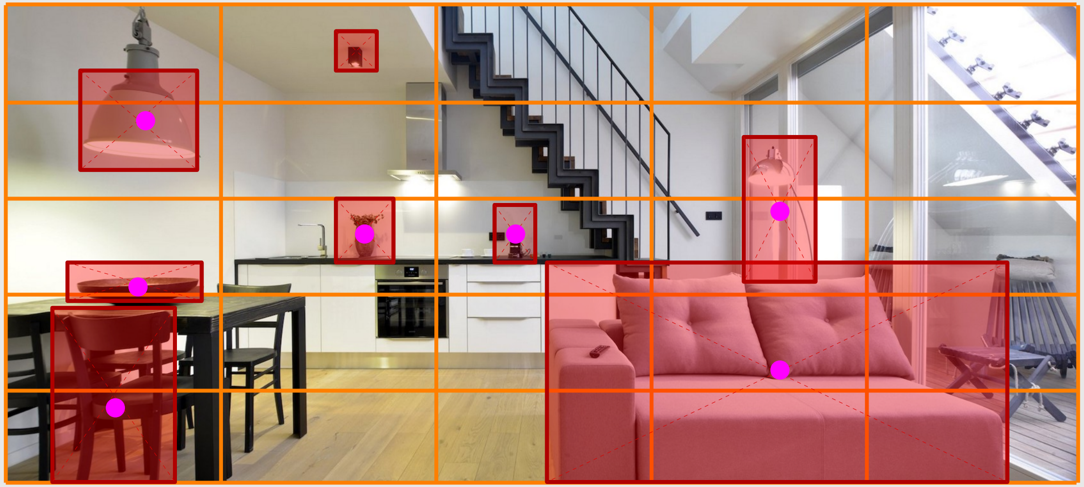
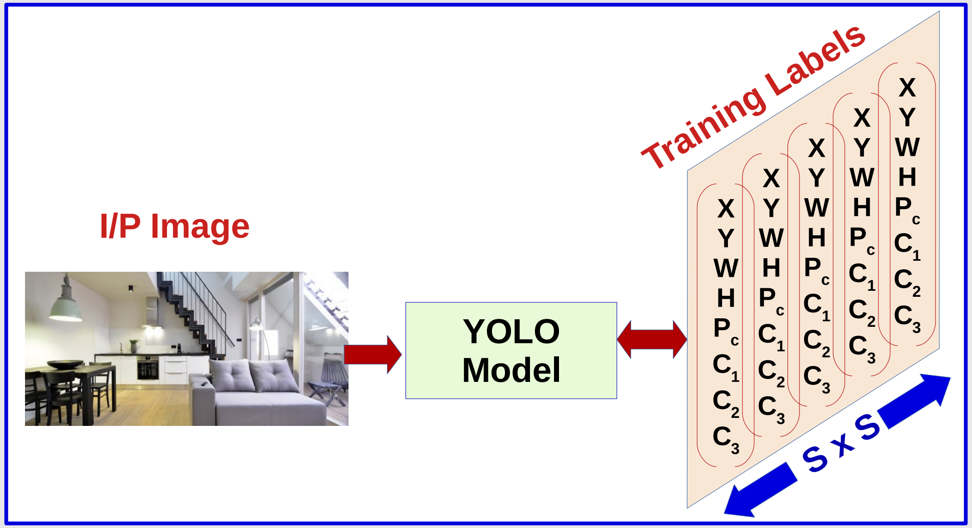
Review of YOLO Basics

YOLO Concept at a Glance





- Divide Image into GRID of **SxS Cells.**
- Annotate All objects with bounding Box and define Object Class.
- Define the CENTER of Each Bounding Box,
- Associate each bounding box with CELL which contains its Center

Construct Y vector (Label) for each CELL which contain:

Coord (x,y,W,H) of bounding box,
Class of the Object (One Hot Encoded),
Confidence value that there is an object center in this cell

YOLO Model is trained to Predict:

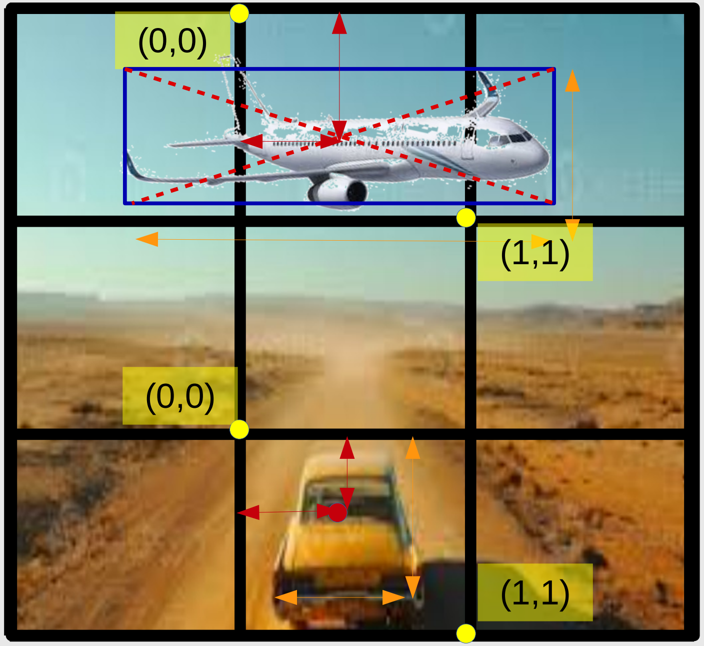
For Each Cell

- Is there an object in this cell (P_c)
- Exact Location of Object (x,y)
- Exact Size of Object (W,H)
- Object Class (C_1, C_2, ..) *[same as Sliding Window]*

Problems:

Same Object may be Predicted in multiple Cells ??
Cell Contains Multiple Objects ??

Encoding of Object Coordinate ([1] Cell Based Encoding)



	X,	*Y,*	*W,*	*H*
Plane:	***0.4,***	***0.7,***	***1.9,***	***0.6***
Car:	***0.5,***	***0.4,***	***0.5,***	***0.7***

Object Center

Image is divides into Grid Cells.

Each Grid box is associated with ONE Object Center ***(One Anchor)***

For Each Gird Cell,

(0,0) is the top left corner

(1,1) is the bottom right corner

X,Y are relative to top left corner **of Cell**

X,Y values ranges from (0 to 1)

Width and height may exceed “1”

Encoding of Object Coordinate ([2] Image Based Encoding)

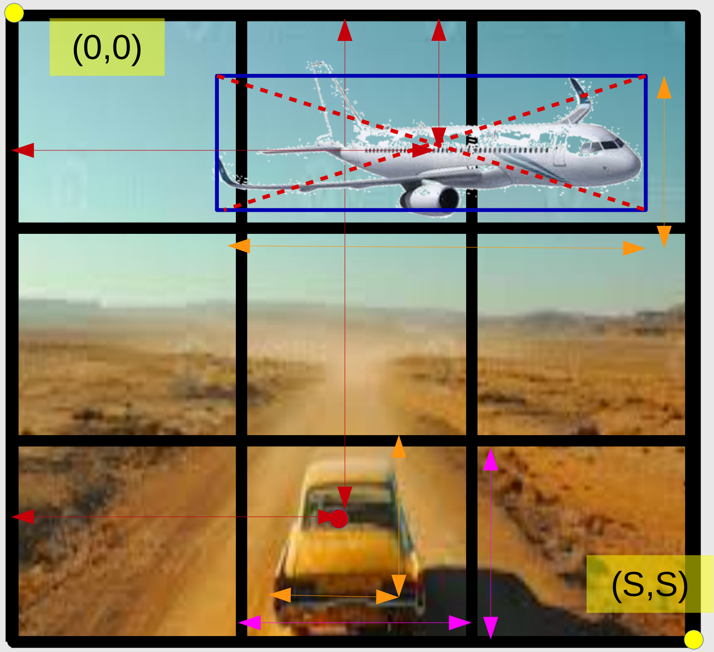


*Image is divided into S*S Cells*
(e.g. 3x3 Cells)

	X,	Y,	W,	H
Plane:	*1.8,*	*0.7,*	*1.9,*	*0.6*
Car:	*1.5,*	*2.4,*	*0.5,*	*0.7*

Image is divides into Grid cells.

Each Grid box is associated with

ONE Object Center *(One Anchor)*

Using Cells Unit **=1**

(0,0) is the top left corner of the IMAGE

(S,S) is the bottom right corner of the Image

X,Y are relative to top left corner **of Image**

X,Y values ranges from (0.x to S)

Width and height ranges from (0.x to S)

Anchor Boxes
(For Multiple Objects Detection)

One Anchor Box

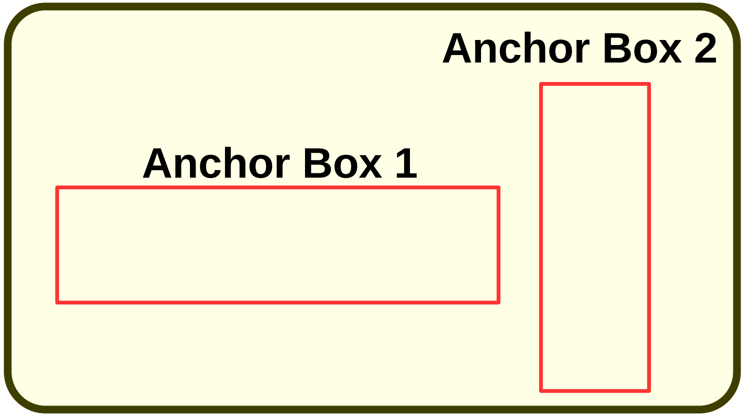
Remember: One Grid Cell Contains One Object at Most (No Overlapped Objects)

Max Number of Detected Objects = Cell Conut

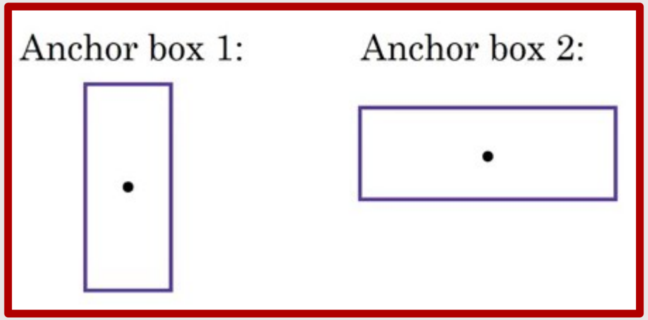
Each Object in training image is assigned to grid cell that contains that object's midpoint

What if there are multiple objects in a single grid?

That can so often be the case in reality in Many Cases Such as:
- Overlapped Objects in Same grid
- Large grid size and small objects
That leads to the concept of anchor boxes

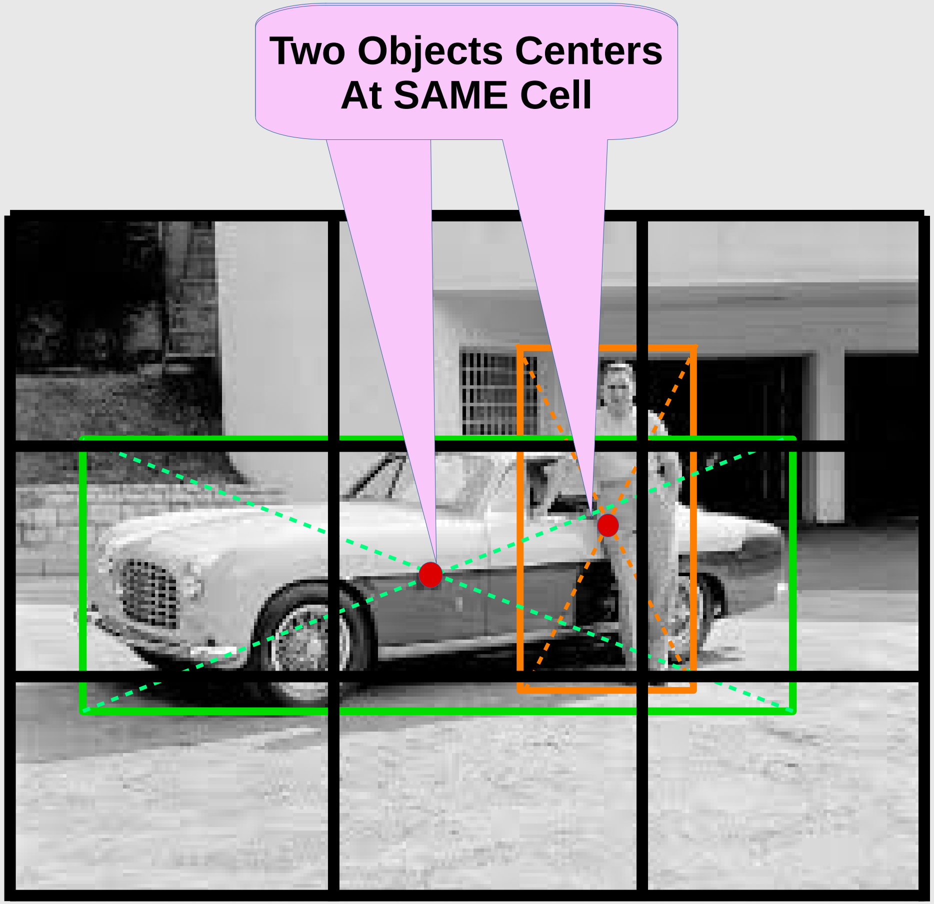


Multiple Anchor Boxes (for Overlapped Objects)

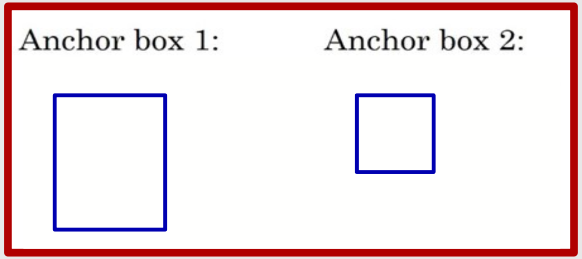


The objects are assigned to the anchor boxes based on:
The similarity of the bounding boxes and the anchor box shape.

Since the shape of **anchor box 1 is similar to the bounding box for the person**,
the Person will be assigned to anchor box 1
and
the **car will be assigned to anchor box 2**.

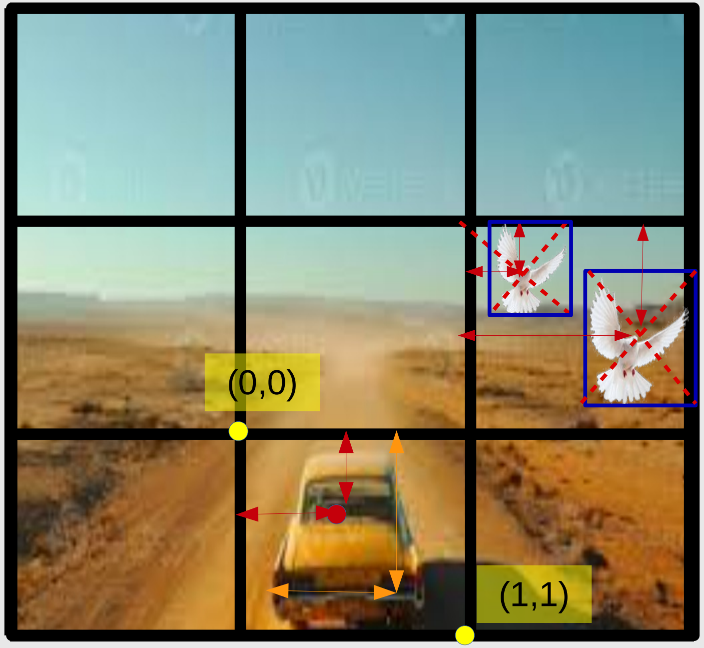


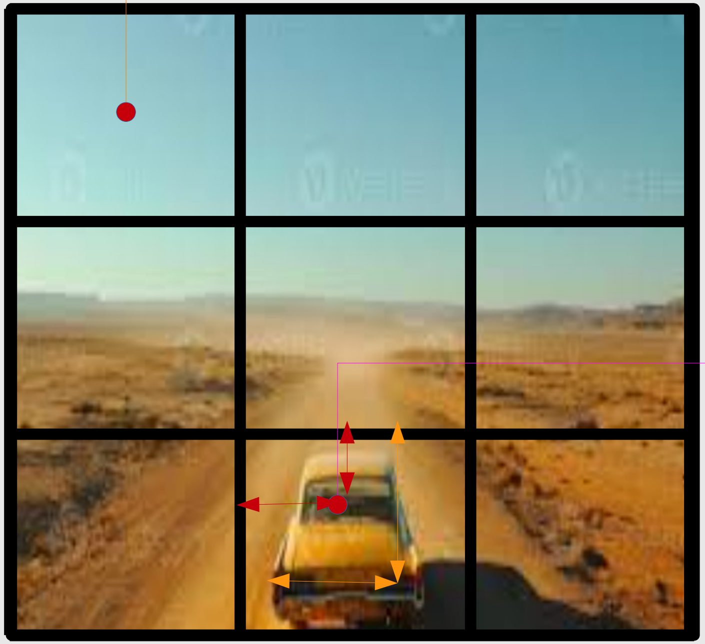
Multiple Anchor Boxes (for Small Objects)



- Select Anchor Box <u>Similar</u> to Each Object (size and Aspect Ratio)

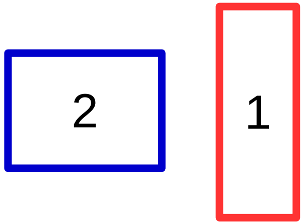
- Each Anchor Box will be associated with (Object **X,Y,W,H**, Object **Class Type**, In Addition to **Confidence** Probability that there Exist an object)





$$\begin{array}{c} 1 \\ \\ 2 \end{array} \quad \left[\begin{array}{c} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{array}\right] \quad \left[\begin{array}{c} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{array}\right] \quad \left[\begin{array}{c} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{array}\right]$$

Two Anchor Boxes



1 - pedestrian
2 - car
3 - motorcycle

y is $3 \times 3 \times 2 \times 8$

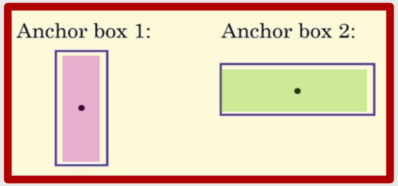
Num. of Boxes In the Grid

Num. of Anchor Boxes

P_c, b_x, b_y, b_w, b_h, + Num. of Classes

What if Object is Assigned to Wrong Anchor Box?

P_c	1	1	1	1	0	0
X	…	…	…	…	…	…
Y	…	…	…	…	…	…
W	0.4	0.5	0.6	2.8	…	…
H	3.1	2.3	2.9	.25	…	…
C_1	…	…	…	…	…	…
C_2	…	…	…	…	…	…
C_3	…	…	…	…	…	…
P_c	0	0	0	0	1	1
X	…	…	…	…	…	…
Y	…	…	…	…	…	…
W	…	…	…	…	2.5	3.4
H	…	…	…	…	0.3	.35
C_1	…	…	…	…	…	…
C_2	…	…	…	…	…	…
C_3	…	…	…	…	…	…

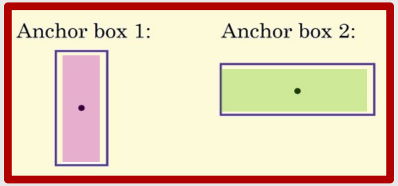


Wrong Anchor

Training Data is NOT Consistent → ***Less Accuracy***

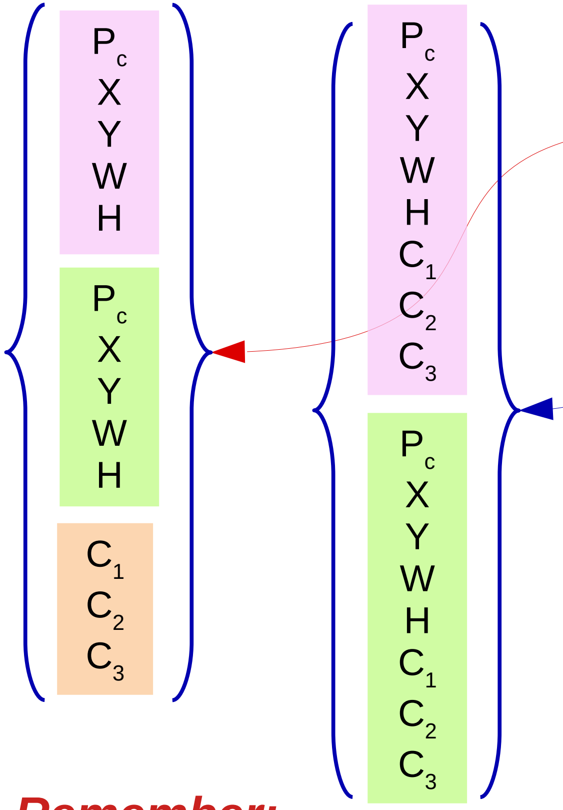
What if Object is Assigned to Wrong Anchor Box?

P_c	1	1	1	0	0	0
X	…	…	…	…	…	…
Y	…	…	…	…	…	…
W	0.4	0.5	0.6	…	…	…
H	3.1	2.3	2.9	…	…	…
C_1	…	…	…	…	…	…
C_2	…	…	…	…	…	…
C_3	…	…	…	…	…	…
P_c	0	0	0	1	1	1
X	…	…	…	…	…	…
Y	…	…	…	…	…	…
W	…	…	…	2.8	2.5	3.4
H	…	…	…	.25	0.3	.35
C_1	…	…	…	…	…	…
C_2	…	…	…	…	…	…
C_3	…	…	…	…	…	…



Proper Anchor **Training Data is Consistent** → *High Accuracy*

Multiple Anchors, ONE Object / Cell



For SxS Cells, M Anchors , N Classes

One Object per Cell

Total Size = [S X S] * [M * 5 + N]

Multiple Objects per Cell

Total Size = [S X S] * M * [5 + N]

Example:

S=13 , **M**=6 Anchors, **N**=100 Class

One Object per Cell

Total Size = 13 * 13 * [6 * 5 + 100] = **21970**

1 : 4.84

Multiple Objects per Cell

Total Size = 13 * 13 * 6 * [5 + 100] = **106470**

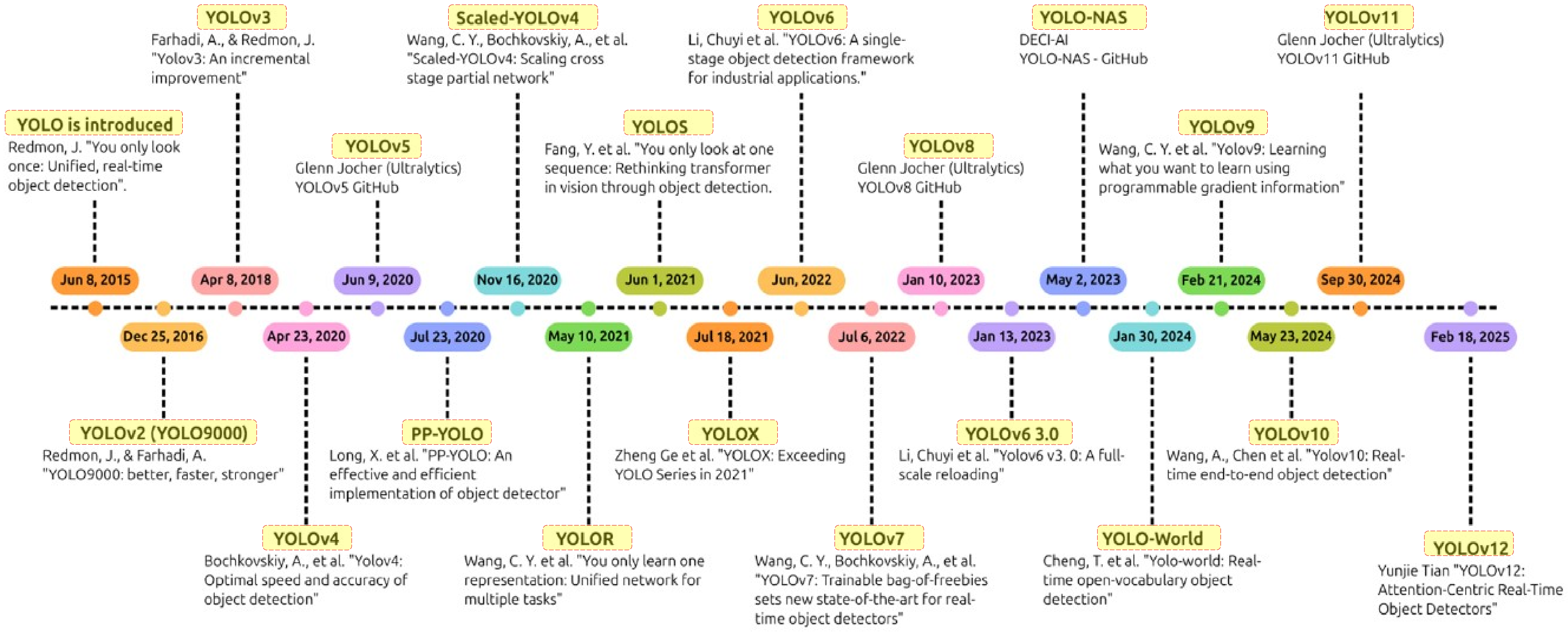
Remember:

ONE Vector / Cell

YOLO Versions (till March 2026)

YOLO,
YOLOv2,
YOLOv3,
YOLOv4,
YOLOv5,
YOLOV6,
YOLOV7,
YOLOv8,
YOLO-NAS,
YOLO-World,
YOLOv9,
YOLOv10,
YOLOv11,
YOLOv12, and
YOLO 2026

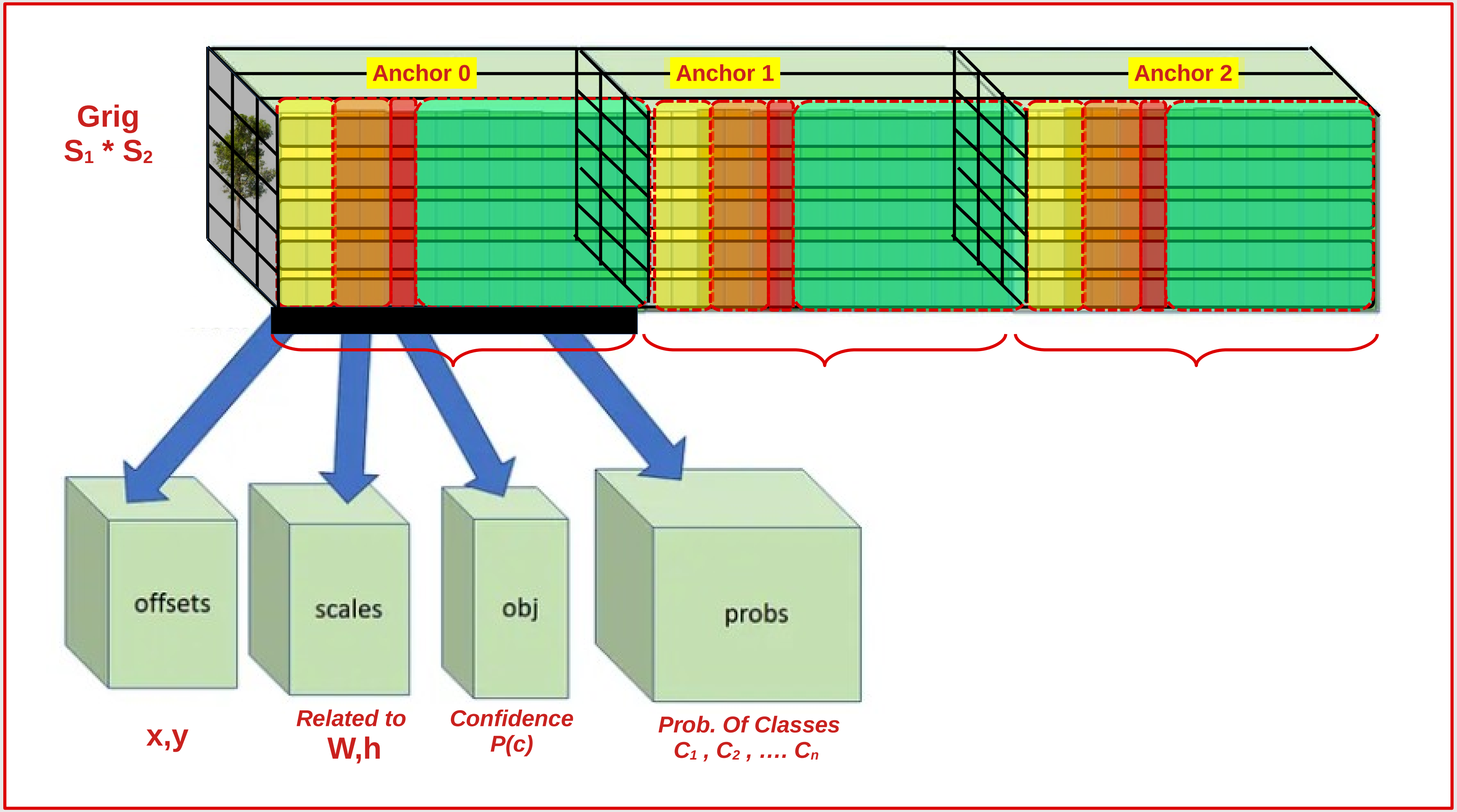
YOLO Versions (till Sept 2025)



From: Analyzing the enhancement of CNN-YOLO and transformer based architectures for real-time animal detection in complex ecological environments
Ali Raza1, Fareeha Hanif2 & Heba Abdelgader Mohammed. Scientific Reports | (2025) 15:39142

Example of YOLO Architecture

YOLO V2 and YOLO V3 output layer.



YOLO
Loss Function

Regression Loss (YOLO Ver 1.0)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Loss of ONE Cell , One Anchor Box

$$\text{Localization Loss} = \begin{pmatrix} (x-\hat{x})^2+(y-\hat{y})^2+(\sqrt{w}-\sqrt{\hat{w}})^2+(\sqrt{h}-\sqrt{\hat{h}})^2 & \Rightarrow & \text{If there is an object} \\ \text{zero} & \Rightarrow & \text{If there is NO object} \end{pmatrix}$$

$$\text{Objectness Loss} = \begin{pmatrix} (C-\hat{C})^2 & \Rightarrow & \text{If there is an object} \\ (C-\hat{C})^2 & \Rightarrow & \text{If there is NO object} \end{pmatrix} \quad \text{Same}$$

$C = \varnothing$

$$\text{Classification Loss} = \begin{pmatrix} \sum_{c \in \text{all classes}} (P(c)-\hat{P}(c))^2 & \Rightarrow & \text{If there is an object} \\ \text{zero} & \Rightarrow & \text{If there is NO object} \end{pmatrix}$$

Predicted	Ground Truth
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$
$p(\hat{c}_2)$	$p(c_2)$
…	…
…	…
$p(\hat{C}_n)$	$p(C_n)$

For ONE Cell

Regression Loss (YOLO Ver 1.0)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Total Loss of ONE Cell , One Anchor Box

$$Loss\ of\ ONE\ Cell = \begin{pmatrix} L^{Obj}\left[(x-\hat{x})^2+(y-\hat{y})^2+(\sqrt{w}-\sqrt{\hat{w}})^2+(\sqrt{h}-\sqrt{\hat{h}})^2\right] & + \\ (C-\hat{C})^2 & + \\ L^{Obj}\left[\sum_{c\in all\ classes}(P(c)-\hat{P}(c))^2\right] & + \end{pmatrix}$$

No Weights

$$Where \begin{pmatrix} L^{Obj}=1 & \text{if there is an object} \\ L^{Obj}=0 & \text{if there is NO object} \\ L^{NoObj}=1 & \text{if there is NO object} \end{pmatrix}$$

Predicted	Ground Truth
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$
$p(\hat{c}_2)$	$p(c_2)$
...	...
...	...
$p(\hat{C}_n)$	$p(C_n)$

For ONE Cell

Regression Loss (YOLO Ver 1.0)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Loss of ONE Cell , One Anchor Box

$$\text{Loss of ONE Cell} = \begin{pmatrix} L^{Obj}\left[(x-\hat{x})^2 + (y-\hat{y})^2 + (\sqrt{w}-\sqrt{\hat{w}})^2 + (\sqrt{h}-\sqrt{\hat{h}})^2\right] & + \\ (C-\hat{C})^2 & + \\ L^{Obj}\left[\sum_{c \in \text{all classes}} (P(c)-\hat{P}(c))^2\right] & + \end{pmatrix}$$

No Weights

Predicted	Ground Truth
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$

*Instead of predicting the **width** and **height** directly,*
*Yolo **predicts the square roots** to account for **deviations in small numbers being more significant than the same deviation in big numbers**.*

The square root mapping is used to expand smaller numbers,
e.g. any number in [0,0.25] gets mapped to [0,0.5].

$$\text{Where} \begin{pmatrix} L^{Obj} = 1 & \text{if there is an object} \\ L^{Obj} = 0 & \text{if there is NO object} \\ L^{NoObj} = 1 & \text{if there is NO object} \end{pmatrix}$$

Regression Loss (YOLO Ver 1.0)
(Adding Weights)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Loss of ONE Cell , One Anchor Box

$$Loss\ of\ ONE\ Cell = \left(\begin{array}{l} L^{Obj}[(x-\hat{x})^2+(y-\hat{y})^2+(\sqrt{w}-\sqrt{\hat{w}})^2+(\sqrt{h}-\sqrt{\hat{h}})^2] \quad + \\ (C-\hat{C})^2 \quad + \\ L^{Obj}[\sum_{c \in all\ classes} (P(c)-\hat{P}(c))^2] \end{array} \right)$$

No Weights

$$Loss\ of\ ONE\ Cell = \left(\begin{array}{l} \lambda_{Coord}[\ L^{Obj}[(x-\hat{x})^2+(y-\hat{y})^2+(\sqrt{w}-\sqrt{\hat{w}})^2+(\sqrt{h}-\sqrt{\hat{h}})^2]] \quad + \\ L^{Obj}[(C-\hat{C})^2] \quad + \\ \lambda_{NoObj}[\ L^{NoObj}[(C-\hat{C})^2]] \quad + \\ L^{Obj}[\sum_{c \in all\ classes} (P(c)-\hat{P}(c))^2] \end{array} \right)$$

Where
- $L^{Obj} = 1$ if there is an object
- $L^{Obj} = 0$ if there is NO object
- $L^{NoObj} = 1$ if there is NO object

Where
- λ_{Coord} Weight for Localization loss
- λ_{NoObj} Weight for Objectness loss if there is NO object

Predicted	*Ground Truth*
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$
$p(\hat{c}_2)$	$p(c_2)$
...	...
...	...
$p(\hat{C}_n)$	$p(C_n)$

For ONE Cell

Regression Loss (YOLO Ver 1.0)
(Adding Weights)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Loss of ONE Cell , One Anchor Box

$$\text{Loss of ONE Cell} = \left(\begin{array}{l} \lambda_{Coord} \left[L^{Obj} \left[(x-\hat{x})^2 + (y-\hat{y})^2 + (\sqrt{w}-\sqrt{\hat{w}})^2 + (\sqrt{h}-\sqrt{\hat{h}})^2 \right] \right] + \\ L^{Obj} \left[(C-\hat{C})^2 \right] + \\ \lambda_{NoObj} \left[L^{NoObj} \left[(C-\hat{C})^2 \right] \right] + \\ L^{Obj} \left[\sum_{c \in all\ classes} (P(c)-\hat{P}(c))^2 \right] \end{array} \right)$$

Predicted	Ground Truth
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$
$\hat{c}_2)$	$p(c_2)$
..	...
..	...
$\hat{C}_n)$	$p(C_n)$

or ONE Cell

Note : *Most of the grids usually have no objects .*
This part of equation will get lot of attention
let $\lambda_{NoObject}$ *has a very small value* $(e.g.\ \mathbf{0.5})$ in *order the* model will NOT be biased about No Object .

let $\lambda_{coordinate}$ *has higher value that* $\lambda_{NoObject}$ $(\mathbf{5.0}\ \ vs\ \ \mathbf{0.5})$
to force the model to focus more about predicting correct objects coordinates (bounding boxes)

Where
λ_{Coord} **Weight for Localization loss**
λ_{NoObj} **Weight for Objectness loss** if there is NO object

Regression Loss (YOLO Ver 1.0)

(All Cells)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Loss of ALL Cells , One Anchor Box

$$Loss\ of\ ALL\ Cells\ (S^2) = \left| \begin{array}{l} \sum_{i=1}^{S^2} \lambda_{Coord} \left[L_i^{Obj} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 + (\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] \right] + \\ \sum_{i=1}^{S^2} L_i^{Obj} \left[(C_i - \hat{C}_i)^2 \right] + \\ \sum_{i=1}^{S^2} \lambda_{NoObj} \left[L_i^{NoObj} \left[(C_i - \hat{C}_i)^2 \right] \right] + \\ \sum_{i=1}^{S^2} L_i^{Obj} \left[\sum_{c \in all\ classes} (P_i(c) - \hat{P}_i(c))^2 \right] \end{array} \right|$$

$$Where \left| \begin{array}{ll} L^{Obj} = 1 & \text{if there is an object} \\ L^{Obj} = 0 & \text{if there is NO object} \\ L^{NoObj} = 1 & \text{if there is NO object} \end{array} \right|$$

$$Where \left| \begin{array}{ll} \lambda_{Coord} & \text{Weight for Localization loss} \\ \lambda_{NoObj} & \text{Weight for Objectness loss if there is NO object} \end{array} \right|$$

Predicted	*Ground Truth*
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$
$p(\hat{c}_2)$	$p(c_2)$
…	…
…	…
$p(\hat{C}_n)$	$p(C_n)$

For ONE Cell

Regression Loss (YOLO Ver 1.0)

(All Cells - All Anchor Boxes)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

Loss of ALL Cells , "B" Anchor Boxes ,ONE Object per Cell

$$\text{Loss of ALL Cells}\,(S^2)\;\text{All B Anchors}\,(B) = \begin{pmatrix} \sum_{i=1}^{S^2}\sum_{j=1}^{B} \lambda_{Coord}\left[\, L_{ij}^{Obj}\left[(x_i-\hat{x}_i)^2+(y_i-\hat{y}_i)^2+(\sqrt{w_i}-\sqrt{\hat{w}_i})^2+(\sqrt{h_i}-\sqrt{\hat{h}_i})^2\right]\right] + \\ \sum_{i=1}^{S^2}\sum_{j=1}^{B} L_{ij}^{Obj}\left[(C_i-\hat{C}_i)^2\right] + \\ \sum_{i=1}^{S^2}\sum_{j=1}^{B} \lambda_{NoObj}\left[\, L_{ij}^{NoObj}\left[(C_i-\hat{C}_i)^2\right]\right] + \\ \sum_{i=1}^{S^2} L_i^{Obj}\left[\sum_{c\in \text{all classes}} (P_i(c)-\hat{P}_i(c))^2\right] \end{pmatrix}$$

$$\text{Where} \begin{pmatrix} L^{Obj}=1 & \text{if there is an object} \\ L^{Obj}=0 & \text{if there is NO object} \\ L^{NoObj}=1 & \text{if there is NO object} \end{pmatrix}$$

$$\text{Where} \begin{pmatrix} \lambda_{Coord} & \text{Weight for Localization loss} \\ \lambda_{NoObj} & \text{Weight for Objectness loss if there is NO object} \end{pmatrix}$$

Predicted	Ground Truth
$\begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{w} \\ \hat{h} \\ \hat{C} \\ \hat{x} \\ \hat{y} \\ \hat{w} \\ \hat{h} \\ \hat{C} \\ p(\hat{c}_1) \\ p(\hat{c}_2) \\ \dots \\ \dots \\ p(\hat{C}_n) \end{pmatrix}$	$\begin{pmatrix} x \\ y \\ w \\ h \\ C \\ x \\ y \\ w \\ h \\ C \\ p(c_1) \\ p(c_2) \\ \dots \\ \dots \\ p(C_n) \end{pmatrix}$

For ONE Cell, 2 Anchor Boxes

Solving Problem of Objectness Loss
(YOLO Ver 2.0 and YOLO Ver 3.0)

Objectness Loss as Regression (as before)

$$\sum_{i=1}^{S^2}\sum_{j=1}^{B} L_{ij}^{Obj}\left[\left(C_i-\hat{C}_i\right)^2\right]$$

$$\sum_{i=1}^{S^2}\sum_{j=1}^{B} \lambda_{NoObj}\left[\, L_{ij}^{NoObj}\left[\left(C_i-\hat{C}_i\right)^2\right]\right]$$

C_i is either 0 or 1

$\hat{C}_i$ is in range from 0 to 1

$\left(C_i-\hat{C}_i\right)^2$ is $fraction^2$

Very small value

Predicted	Ground Truth
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$\hat{x}$	x
$\hat{y}$	y
$\hat{w}$	w
$\hat{h}$	h
$\hat{C}$	C
$p(\hat{c}_1)$	$p(c_1)$
$p(\hat{c}_2)$	$p(c_2)$
...	...
...	...
$p(\hat{C}_n)$	$p(C_n)$

For ONE Cell, 2 Anchor Boxes

Using Cross Entropy Loss for Objectness Loss
(YOLO Ver 2.0 and YOLO Ver 3.0)

Objectness Loss as Regression (as before)

$$\sum_{i=1}^{S^2}\sum_{j=1}^{B} L_{ij}^{Obj}\left[\left(C_i - \hat{C}_i\right)^2\right]$$

$$\sum_{i=1}^{S^2}\sum_{j=1}^{B} \lambda_{NoObj}\left[\ L_{ij}^{NoObj}\left[\left(C_i - \hat{C}_i\right)^2\right]\right]$$

C_i is either 0 or 1

$\hat{C}_i$ is in range from 0 to 1

$\left(C_i - \hat{C}_i\right)^2$ is $fraction^2$

Very small value

Objectness Loss as Cross Entropy

$$\sum_{i=1}^{S^2}\sum_{j=1}^{B} L_{ij}^{Obj}\left[\hat{C}_i \log\left(C_i\right) + \left(1-\hat{C}_i\right)\log\left(1-C_i\right)\right]$$

$$\sum_{i=1}^{S^2}\sum_{j=1}^{B} \lambda_{NoObj}\left[\ L_{ij}^{NoObj}\left[\hat{C}_i \log\left(C_i\right) + \left(1-\hat{C}_i\right)\log\left(1-C_i\right)\right]\right]$$

Predicted	Ground Truth
$\begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{w} \\ \hat{h} \\ \hat{C} \\ \hat{x} \\ \hat{y} \\ \hat{w} \\ \hat{h} \\ \hat{C} \\ p(\hat{c}_1) \\ p(\hat{c}_2) \\ \cdots \\ \cdots \\ p(\hat{C}_n) \end{pmatrix}$	$\begin{pmatrix} x \\ y \\ w \\ h \\ C \\ x \\ y \\ w \\ h \\ C \\ p(c_1) \\ p(c_2) \\ \cdots \\ \cdots \\ p(C_n) \end{pmatrix}$

For ONE Cell,
2 Anchor Boxes

Using Cross Entropy Loss for Classification Loss
(YOLO Ver 2.0 and YOLO Ver 3.0)

Classification Loss as Regression

$$\sum_{i=1}^{S^2} L_i^{Obj} \Big[\sum_{c \in all\ classes} \big(P_i(c) - \hat{P}_i(c)\big)^2 \Big]$$

$P_i(c)$ is either 0 or 1

$\hat{P}_i(c)$ is in range from 0 to 1

$\big(P_i(c) - \hat{P}_i(c)\big)^2$ is $fraction^2$

Very small value

Classification Loss as Cross Entropy

$$\sum_{i=1}^{S^2} L_i^{Obj} \Big[\sum_{c \in all\ classes} \big[\hat{P}_i(c) \log\big(P_i(c)\big) + \big(1 - \hat{P}_i(c)\big) \log\big(1 - P_i(c)\big)\big] \Big]$$

Predicted	Ground Truth
$\begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{w} \\ \hat{h} \\ \hat{C} \\ \hat{x} \\ \hat{y} \\ \hat{w} \\ \hat{h} \\ \hat{C} \\ p(\hat{c}_1) \\ p(\hat{c}_2) \\ \ldots \\ \ldots \\ p(\hat{C}_n) \end{pmatrix}$	$\begin{pmatrix} x \\ y \\ w \\ h \\ C \\ x \\ y \\ w \\ h \\ C \\ p(c_1) \\ p(c_2) \\ \ldots \\ \ldots \\ p(C_n) \end{pmatrix}$

For ONE Cell,
2 Anchor Boxes

Regression Loss and Cross Entropy Loss (YOLO Ver 2.0 and 3.0)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

$$\text{Loss of ALL Cells, All Anchors} = \left(\begin{array}{l} \sum_{i=1}^{S^2} \sum_{j=1}^{B} \lambda_{Coord} \left[L_{ij}^{Obj} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 + (\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] \right] + \\ \sum_{i=1}^{S^2} \sum_{j=1}^{B} L_{ij}^{Obj} \left[\hat{C}_i \log(C_i) + (1 - \hat{C}_i) \log(1 - C_i) \right] + \\ \sum_{i=1}^{S^2} \sum_{j=1}^{B} \lambda_{NoObj} \left[L_{ij}^{NoObj} \left[\hat{C}_i \log(C_i) + (1 - \hat{C}_i) \log(1 - C_i) \right] \right] + \\ \sum_{i=1}^{S^2} L_i^{Obj} \left[\sum_{c \in \text{all classes}} \left[\hat{P}_i(c) \log(P_i(c)) + (1 - \hat{P}_i(c)) \log(1 - P_i(c)) \right] \right] \end{array} \right)$$

$$\text{Where} \left(\begin{array}{ll} L^{Obj} = 1 & \text{if there is an object} \\ L^{Obj} = 0 & \text{if there is NO object} \\ L^{NoObj} = 1 & \text{if there is NO object} \end{array} \right)$$

$$\text{Where} \left(\begin{array}{ll} \lambda_{Coord} & \text{Weight for Localization loss} \\ \lambda_{NoObj} & \text{Weight for Objectness loss if there is NO object} \end{array} \right)$$

Problem of Regression Loss (for Localization)

$$\text{Localization Loss} = \begin{pmatrix} (x-\hat{x})^2 + (y-\hat{y})^2 + \left(\sqrt{w}-\sqrt{\hat{w}}\right)^2 + \left(\sqrt{h}-\sqrt{\hat{h}}\right)^2 & \Rightarrow & \textit{If there is an object} \\ \textit{zero} & \Rightarrow & \textit{If there is NO object} \end{pmatrix}$$

Note:
Localization Loss May take any value
Objectness Loss is less than 1
Classification Loss is less than 1

Localization Loss will dominate even with weighting factors

IOU instead of Regression Loss (for Localization)

$$Localization\ Loss = \begin{pmatrix} (x-\hat{x})^2+(y-\hat{y})^2+(\sqrt{w}-\sqrt{\hat{w}})^2+(\sqrt{h}-\sqrt{\hat{h}})^2 & \Rightarrow & If\ there\ is\ an\ object \\ zero & \Rightarrow & If\ there\ is\ NO\ object \end{pmatrix}$$

Note:
Localization Loss May take any value
Objectness Loss is less than 1
Classification Loss is less than 1

Localization Loss will dominate even with weighting factors

Solution:
Use **IOU** as a measure of **Similarity** between Predicted Bounding Box and Ground Truth Bounding Box

Loss = 1 - IOU

IOU instead of Regression Loss (for Localization)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

$$\text{Loss of ALL Cells All Anchors} = \left(\begin{array}{l} \sum_{i=1}^{S^2} \sum_{j=1}^{B} L_{ij}^{Obj} \left[1 - IoU_{predicted\ ij}^{Ground\ Truth} \right] + \\ \sum_{i=1}^{S^2} \sum_{j=1}^{B} L_{ij}^{Obj} \left[\hat{C}_i \log(C_i) + (1-\hat{C}_i)\log(1-C_i) \right] + \\ \sum_{i=1}^{S^2} \sum_{j=1}^{B} \lambda_{NoObj} \left[L_{ij}^{NoObj} \left[\hat{C}_i \log(C_i) + (1-\hat{C}_i)\log(1-C_i) \right] \right] + \\ \sum_{i=1}^{S^2} L_i^{Obj} \left[\sum_{c \in all\ classes} \left[\hat{P}_i(c) \log(P_i(c)) + (1-\hat{P}_i(c)) \log(1-P_i(c)) \right] \right] \end{array} \right)$$

$$\text{Where} \left(\begin{array}{ll} L^{Obj} = 1 & \text{if there is an object} \\ L^{Obj} = 0 & \text{if there is NO object} \\ L^{NoObj} = 1 & \text{if there is NO object} \end{array} \right)$$

Which Variant of IoU ??

- Basic IoU
- GIoU (Generalized IoU)
- DIoU (Distance IoU)
- CIoU (Complete IoU)
- EIoU (Efficient IoU)
- Focal-EIoU
- SIoU (SCYLLA-IoU)
- WIoU (Wise-IoU)

Which Variant of IoU

Starting from YOLOv4, researchers started focusing more on the IoU-based losses, as it was a better estimate of bounding box localization accuracy

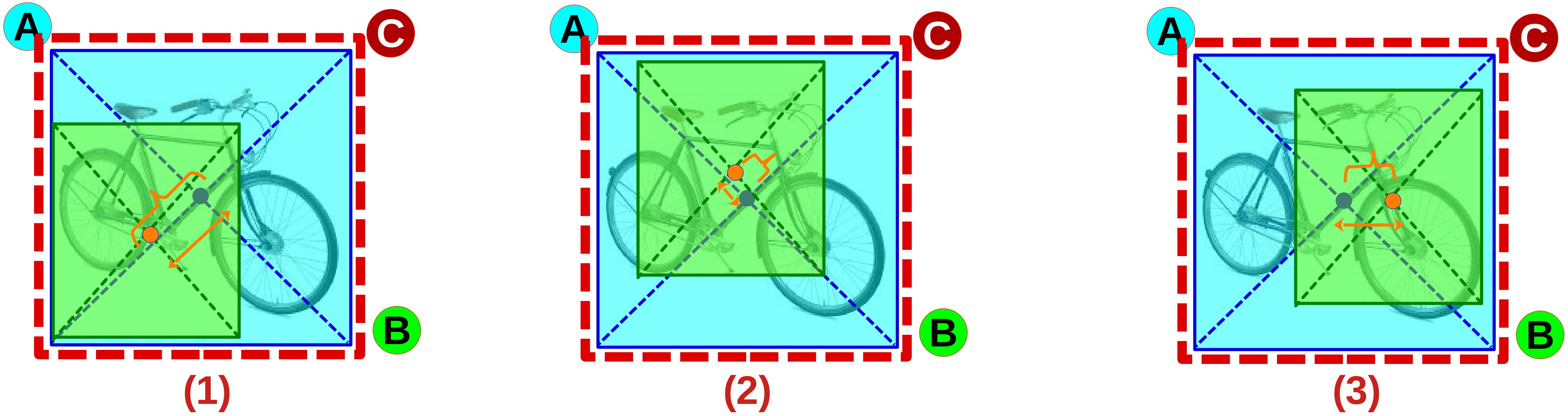
YOLOv5, **YOLOv4**, **YOLOR** and **YOLOv7:**
Authors used **CIoU** (Complete IOU) loss for their bounding box regression loss.

YOLOX:
Authors decided to go for **Normal IoU** loss,

YOLOv6:
Authors used **SIoU/GIoU** as regression loss and VariFocal loss for classification task,

YOLOv8:
Authors used both **CIoU** and Distribution Focal Loss (**DFL**) loss functions for their bounding box regression loss.

Review: DIoU (Distance IoU) Loss Function



Where :

d^2 : *Squared distance between the* ***two centers***

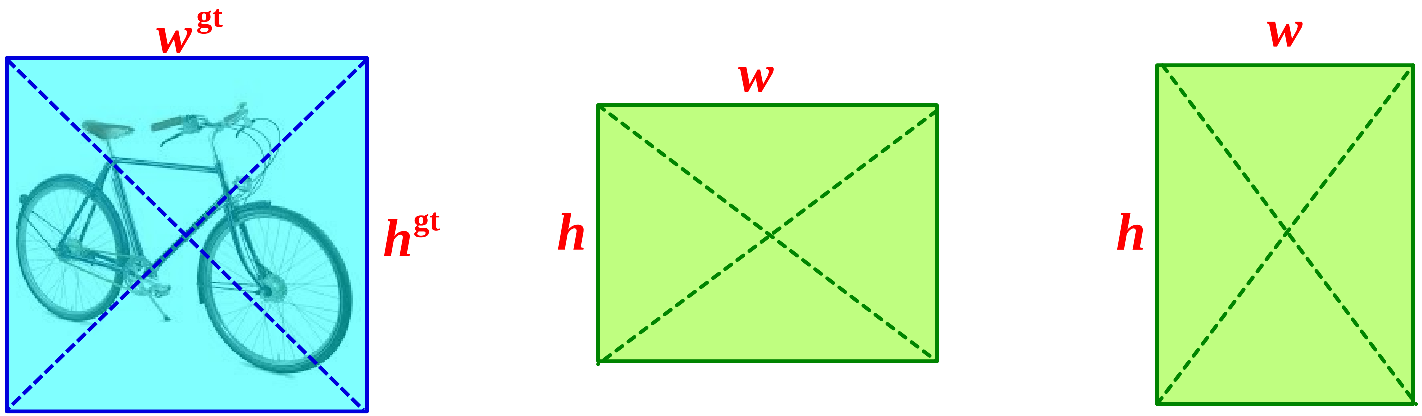
C^2 : *Squared length of* ***diagonal*** *of* **C**

$$IoU = \left| \frac{A \cap B}{A \cup B} \right|$$

$$D\,IOU = IoU - \frac{d^2}{C^2}$$

$$Loss_{D\,IOU} = 1 - IoU - \frac{d^2}{C^2}$$

Review: CIoU (Complete IoU) Loss Function
(Measure of Orientation)



Ground Truth

$$angle = \arctan \frac{w^{gt}}{h^{gt}}$$

$$angle = \arctan \frac{w}{h}$$

$$angle = \arctan \frac{w}{h}$$

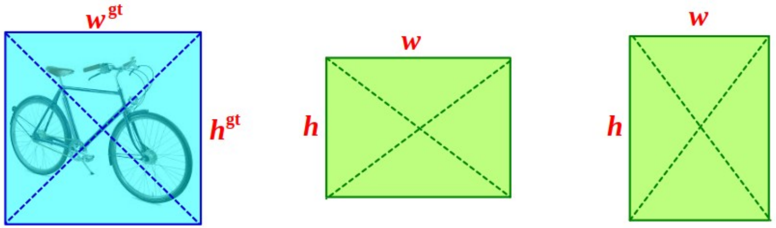
angle range $(0 \quad \text{to} \quad \pi/2)$

$$Angle\ Distance = \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h}\right)^2$$

Angle Distance is a measure of ***Orientation***

$$Normalized\ Angle\ Distance = \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h}\right)^2 / (\pi/2)^2 = \frac{4}{\pi^2}\left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h}\right)^2$$

Review: CIoU (Complete IoU) Loss Function
(Measure of Orientation)



$$IoU = \left| \frac{A \cap B}{A \cup B} \right|$$

$$D\,IOU = IoU - \frac{d^2}{C^2}$$

$$Orientation\ \nu = \frac{4}{\pi^2}\left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h}\right)^2$$

$$C\,IOU = IoU - \frac{d^2}{C^2} - \alpha\nu$$

Where α *is* ***oreintation*** *weight factor based on IOU* $\qquad \alpha = \frac{\nu}{(1 - IoU) + \nu}$

$$Loss_{C\,IOU} = 1 - IoU + \frac{d^2}{C^2} + \alpha\nu$$

CIoU Based Regression Loss (for Localization)

Loss = [Localization Loss , Objectness Loss , Classification Loss]

$$\text{Loss of ALL Cells, All Anchors} = \left(\begin{array}{l} \sum_{i=1}^{S^2} \sum_{j=1}^{B} L_{ij}^{Obj} \left[1 - IoU_{predicted\ ij}^{Ground\ Truth} + \frac{d^2}{C^2} + \alpha v \right] + \\ \sum_{i=1}^{S^2} \sum_{j=1}^{B} L_{ij}^{Obj} \left[\hat{C}_i \log(C_i) + (1-\hat{C}_i) \log(1-C_i) \right] + \\ \sum_{i=1}^{S^2} \sum_{j=1}^{B} \lambda_{NoObj} \left[L_{ij}^{NoObj} \left[\hat{C}_i \log(C_i) + (1-\hat{C}_i) \log(1-C_i) \right] \right] + \\ \sum_{i=1}^{S^2} L_i^{Obj} \left[\sum_{c \in all\ classes} \left[\hat{P}_i(c) \log(P_i(c)) + (1-\hat{P}_i(c)) \log(1-P_i(c)) \right] \right] \end{array} \right)$$

$$\text{Where} \left(\begin{array}{ll} L^{Obj} = 1 & \text{if there is an object} \\ L^{Obj} = 0 & \text{if there is NO object} \\ L^{NoObj} = 1 & \text{if there is NO object} \end{array} \right)$$

Most Enhancements of YOLO Versions are in Loss Function

Other Ideas to Enhance Loss function ??

Get more ideas from the papers

Loss Types of YOLO Versions (till Sept 2025)

Version	Loss type	Mathematical formulation	Key improvements
YOLOv1	Sum-squared Error	$\mathcal{L} = \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} 1_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2\right] + \left[(y_i - \hat{y}_i)^2\right] + \cdots$	Simple implementation, Poor IoU handling, Struggles with small objects
YOLOv2–v3	Cross-Entropy + IoU-like	$\mathcal{L}_{\text{obj}} = -\sum_i 1_i \log(\hat{p}_i)$	Better objectness modeling, Improved stability, Multi-scale support
YOLOv4	CIoU + BCE	$\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\text{b}, \text{b}^{gt})}{c^2} + \alpha v$	Considers aspect ratio, Better convergence, Center distance penalty
YOLOv5	CIoU + Focal Loss	$\mathcal{L}_{\text{focal}} = \alpha_t (1 - p_t)^\gamma \log(p_t)$	Handles class imbalance, Focuses on hard examples, Auto-anchor tuning
YOLOv6	SIoU Loss	$\mathcal{L}_{\text{SIoU}} = \text{Angle cost} + \text{Shape cost} + \text{IoU}$	Geometric precision, Simplified gradients, Efficient decoupled head
YOLOv7	EIoU Loss	$\mathcal{L}_{\text{EIoU}} = 1 - \text{IoU} + \frac{(x - x^g)^2}{w^g} + \frac{(y - y^g)^2}{h^g}$	Better width-height scaling, Improved box matching
YOLOv8	DFL + CIoU	$\mathcal{L}_{\text{DFL}} = -\sum_i p_i^* \log(p_i)$	Continuous distribution prediction, NMS-free design
YOLOv9	VLR Loss	Vector projection with boundary constraints	Eliminates NMS, Learned object ordering, PGI gradient flow
YOLOv10	Probabilistic Loss	Joint classification-regression likelihoods	End-to-end training, Fewer false positives, Transformer integration
YOLOv11	GELAN Loss	Hybrid geometric + attention-guided IoU	Density awareness, Soft edge handling, SAM integration
YOLOv12	Unified Ranking Loss	Directional attention box matching	Occlusion robustness, Instance ordering, Foundation model fusion

From: Analyzing the enhancement of CNN-YOLO and transformer based architectures for real-time animal detection in complex ecological environments